

Divergence Theorem

Let E be a simple solid region and S is the boundary surface of E with positive orientation. Let $\vec{F}$ be a vector field whose components have continuous first order partial derivatives. Then,

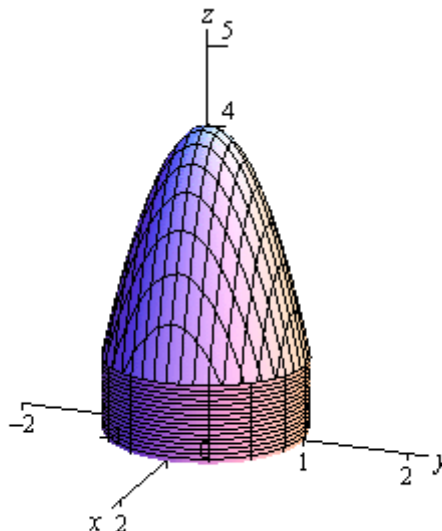
$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_E \operatorname{div} \vec{F} \, dV$$

Let's see an example of how to use this theorem.

Example 1 Use the divergence theorem to evaluate $\iint_S \vec{F} \cdot d\vec{S}$ where

$\vec{F} = xy\vec{i} - \frac{1}{2}y^2\vec{j} + z\vec{k}$ and the surface consists of the three surfaces, $z = 4 - 3x^2 - 3y^2$, $1 \leq z \leq 4$ on the top, $x^2 + y^2 = 1$, $0 \leq z \leq 1$ on the sides and $z = 0$ on the bottom.

Let's start this off with a sketch of the surface.



The region E for the triple integral is then the region enclosed by these surfaces. Note that cylindrical coordinates would be a perfect coordinate system for this region. If we do that here are the limits for the ranges.

$$0 \leq z \leq 4 - 3r^2$$

$$0 \leq r \leq 1$$

$$0 \leq \theta \leq 2\pi$$

We'll also need the divergence of the vector field so let's get that.

$$\operatorname{div} \vec{F} = y - y + 1 = 1$$

The integral is then,

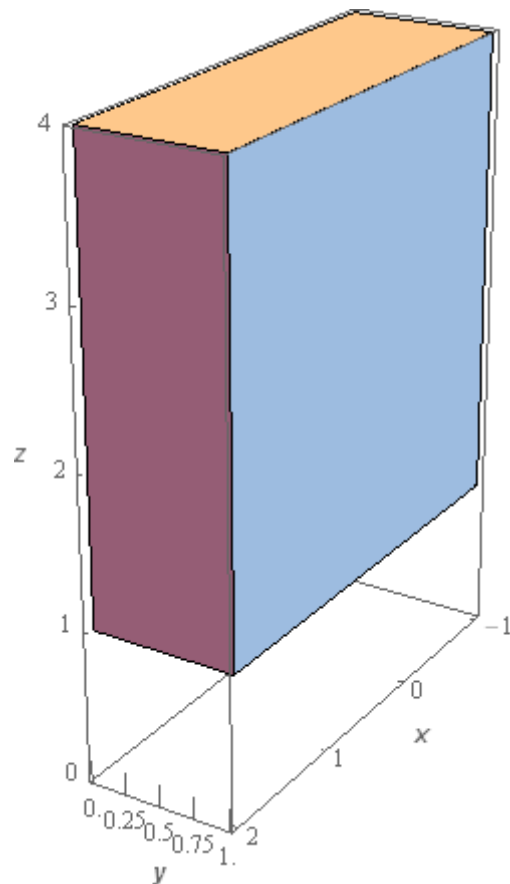
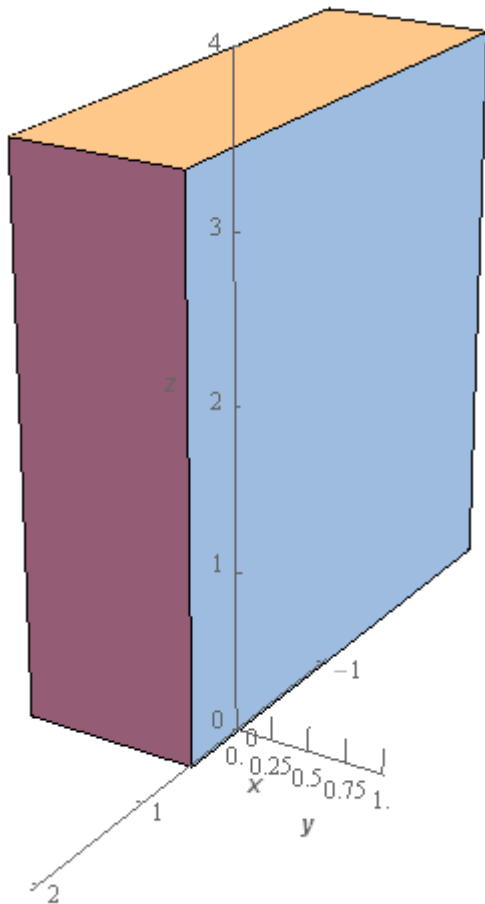
$$\begin{aligned} \iint_S \vec{F} \cdot d\vec{S} &= \iiint_E \operatorname{div} \vec{F} \, dV \\ &= \int_0^{2\pi} \int_0^1 \int_0^{4-3r^2} r \, dz \, dr \, d\theta \\ &= \int_0^{2\pi} \int_0^1 (4r - 3r^3) \, dr \, d\theta \\ &= \int_0^{2\pi} \left(2r^2 - \frac{3}{4}r^4 \right) \Big|_0^1 d\theta \\ &= \int_0^{2\pi} \frac{5}{4} \, d\theta \\ &= \frac{5}{2}\pi \end{aligned}$$

Example for divergence Theorem

2. Use the Divergence Theorem to evaluate $\iint_S \vec{F} \cdot d\vec{S}$ where

$\vec{F} = \sin(\pi x) \vec{i} + zy^3 \vec{j} + (z^2 + 4x) \vec{k}$ and S is the surface of the box with $-1 \leq x \leq 2$, $0 \leq y \leq 1$ and $1 \leq z \leq 4$. Note that all six sides of the box are included in S .

Let's start off with a quick sketch of the surface we are working with in this problem.



We included a sketch with traditional axes and a sketch with a set of "box" axes to help visualize the surface.

Note as well here that because we are including all six sides of the box shown above that the surface does enclose (or is the boundary curve if you want to use that terminology) for the box.

We are going to use the Divergence Theorem in the following direction.

$$\iint_S \vec{F} \cdot d\vec{S} = \iiint_E \operatorname{div} \vec{F} dV$$

where E is just the solid shown in the sketches from Step 1.

E is just a box and the limits defining it where given in the problem statement. The limits for our integral will then be,

$$\begin{aligned} -1 &\leq x \leq 2 \\ 0 &\leq y \leq 1 \\ 1 &\leq z \leq 4 \end{aligned}$$

We'll also need the divergence of the vector field so here is that.

$$\operatorname{div} \vec{F} = \frac{\partial}{\partial x}(\sin(\pi x)) + \frac{\partial}{\partial y}(zy^3) + \frac{\partial}{\partial z}(z^2 + 4x) = \pi \cos(\pi x) + 3zy^2 + 2z$$

Now let's apply the Divergence Theorem to the integral and get it converted to a triple integral.

$$\begin{aligned} \iint_S \vec{F} \cdot d\vec{S} &= \iiint_E \operatorname{div} \vec{F} dV \\ &= \iiint_E \pi \cos(\pi x) + 3zy^2 + 2z dV \\ &= \int_{-1}^2 \int_0^1 \int_1^4 \pi \cos(\pi x) + 3zy^2 + 2z dz dy dx \end{aligned}$$

All we need to do then in evaluate the integral.

$$\begin{aligned}
 \iint_S \vec{F} \cdot d\vec{S} &= \int_{-1}^2 \int_0^1 \int_1^4 \pi \cos(\pi x) + 3zy^2 + 2z \, dz \, dy \, dx \\
 &= \int_{-1}^2 \int_0^1 \left(\pi z \cos(\pi x) + \frac{3}{2} z^2 y^2 + z^2 \right) \Big|_1^4 \, dy \, dx \\
 &= \int_{-1}^2 \int_0^1 3\pi \cos(\pi x) + \frac{45}{2} y^2 + 15 \, dy \, dx \\
 &= \int_{-1}^2 \left(3y\pi \cos(\pi x) + \frac{15}{2} y^3 + 15y \right) \Big|_0^1 \, dx \\
 &= \int_{-1}^2 3\pi \cos(\pi x) + \frac{45}{2} \, dx \\
 &= \left(3 \sin(\pi x) + \frac{45}{2} x \right) \Big|_{-1\pi}^2 = \boxed{\frac{135}{2}}
 \end{aligned}$$